

System Design Specification

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1 Feature 1: Vet Registration

1.1 Sub Features

- Personal Information Submission
- Professional Credential Submission
- Clinic Information Setup
- Account Setup
- Document Verification
- Approval/Rejection
- Profile Activation

1.2 Metadata

Personal Information: VetID, FullName, DateOfBirth, NID, ProfilePhoto

Credentials: LicenseNumber, LicenseExpiry, DegreeCertificate, Specialization, ExperienceYears

Clinic: ClinicName, ClinicAddress, GPSLocation, ServiceRadius, ConsultationFee

Account: Email, Password, PhoneNumber, OTP

Verification: VerificationStatus, VerificationDate, RejectionReason

1.3 Stakeholder Analysis

Stakeholder	Role / Interest	Impact Level
Veterinarian	Primary actor; submits credentials and clinic info to gain platform access	High — directly depends on this feature
Platform Admin	Reviews and approves/rejects vet applications; enforces compliance	High — controls quality of vet pool
Farmer	Indirect beneficiary; relies on verified, trustworthy vets	Medium — trust depends on vet quality
Document Storage Service	Stores credential files (license, degree) securely	Medium — storage failure disrupts registration

1.4 Risk Analysis

Risk	Likelihood	Impact	Mitigation
Fraudulent credential submission	Medium	High	Manual admin verification + API cross-check with licensing boards
OTP delivery failure	Medium	Medium	Retry mechanism + fallback to email OTP
Admin bottleneck in approval	High	Medium	SLA enforcement + automated pre-screening
Data loss during upload	Low	High	Chunked upload + resumable upload support

1.5 Detailed Flow

Step	Action	Metadata Used
1	Vet Registration Form	FullName, DateOfBirth, NID, ProfilePhoto
2	Upload Credentials	LicenseNumber, LicenseExpiry, DegreeCertificate, Specialization, ExperienceYears
3	Add Clinic Information	ClinicName, ClinicAddress, GPSLocation, ServiceRadius, ConsultationFee
4	Account Creation	Email, Password, PhoneNumber, OTP
5	Verification	LicenseNumber, DegreeCertificate, VerificationStatus
6	Approval/Rejection	VerificationStatus, RejectionReason
7	Profile Goes Live	VetID, VerificationStatus

2 Feature 2: Appointment Booking

2.1 Sub Features

- Vet Search
- Vet Profile View
- Slot Selection
- Appointment Request
- Accept/Reject
- Consultation
- Follow-up

2.2 Metadata

Appointment: AppointmentID, FarmerID, VetID, AnimalType, Symptoms, AppointmentDate, AppointmentTime, BookingType, Status

2.3 Stakeholder Analysis

Stakeholder	Role / Interest	Impact Level
Farmer	Initiates booking; searches vet, selects slot, submits request	High — core user of this feature
Veterinarian	Receives and accepts/rejects appointment requests; provides consultation	High — manages schedule and patient load
Platform Admin	Monitors booking activity; handles disputes and cancellations	Medium — oversight and conflict resolution
Notification Service	Sends booking confirmation, reminders, and status updates	Medium — failure affects communication
Calendar / Scheduling Engine	Manages vet availability and prevents double booking	High — core dependency
Payment Gateway	Triggered after booking confirmation for fee collection	Medium — financial dependency

2.4 Risk Analysis

Risk	Likelihood	Impact	Mitigation
Double booking same slot	Medium	High	Optimistic locking on slot reservation + database transaction
Vet no-show after confirmation	Medium	High	Automated reminders + penalty policy for repeated no-shows
Farmer submits incorrect symptoms	High	Medium	Guided symptom input form with predefined options
GPS-based search inaccuracy	Medium	Medium	Fallback to manual area/district search
No available slots for urgent case	Medium	High	Emergency queue with priority booking option

2.5 Detailed Flow

Step	Action	Metadata Used
1	Search Vet	GPSLocation, Specialization, ConsultationFee
2	View Vet Profile	VetID, FullName, Rating, ExperienceYears
3	Select Slot	AppointmentDate, AppointmentTime
4	Submit Request	FarmerID, VetID, AnimalType, Symptoms, BookingType
5	Vet Reviews Request	AppointmentID, Symptoms, AnimalType
6	Accept/Reject	Status
7	Consultation	AppointmentID, BookingType
8	Follow-up Decision	AppointmentID, Status

3 Feature 3: AI Disease Detection

3.1 Sub Features

- Image Upload
- Symptom Submission
- AI Analysis
- Recommendation

3.2 Metadata

DetectionID, FarmerID, AnimalType, AnimalAge, AnimalImage, Symptoms, Prediction, ConfidenceScore, DetectionDate

3.3 Stakeholder Analysis

Stakeholder	Role / Interest	Impact Level
Farmer	Primary user; uploads animal image and symptoms for diagnosis	High — relies on results for animal health
AI/ML Model	Core engine; processes image and symptoms to generate prediction	High — accuracy directly affects outcomes
Veterinarian	Secondary validator; may review AI predictions and provide corrections	Medium — ensures clinical validity
Platform Admin	Monitors model performance and reviews edge case predictions	Medium — responsible for model governance
ML/Data Engineering Team	Trains and updates the AI model; manages dataset quality	High — model accuracy depends on them
Regulatory Authority	Ensures AI health recommendations comply with veterinary regulations	Medium — legal liability for wrong advice

3.4 Risk Analysis

Risk	Likelihood	Impact	Mitigation
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Low-confidence or wrong prediction	High	High	Display ConfidenceScore; suggest vet consultation below threshold
Poor image quality from users	High	Medium	Image quality check before submission; in-app capture guide
Model bias for rare diseases	Medium	High	Regular retraining with diverse datasets; confidence flagging
Farmer over-reliance on AI	High	High	Clear disclaimers; always recommend vet follow-up
Model downtime	Low	High	Fallback to symptom-only decision tree; offline mode
Data leakage of animal health info	Low	Medium	Encrypt DetectionID records; anonymize training data

3.5 Detailed Flow

Step	Action	Metadata Used
1	Upload Image	AnimalImage
2	Enter Symptoms	Symptoms
3	AI Analysis	AnimalImage, Symptoms, AnimalType
4	Generate Prediction	Prediction, ConfidenceScore
5	Recommendation	Prediction, ConfidenceScore

4 Feature 4: Payment System

4.1 Sub Features

- Payment Summary
- Payment Method Selection
- Transaction Processing
- Confirmation
- Payout

4.2 Metadata

TransactionID, AppointmentID, FarmerID, VetID, Amount, PaymentMethod, PaymentStatus, PaymentDate

4.3 Stakeholder Analysis

Stakeholder	Role / Interest	Impact Level
Farmer	Payer; needs a secure and convenient payment method	High — financial exposure is direct
Veterinarian	Payee; depends on timely and accurate payout	High — revenue depends on this feature
Payment Gateway (e.g. bKash, SSLCommerz)	Processes transactions; provides payment APIs	High — unavailability blocks all payments
Platform (as intermediary)	Collects service fee; manages payout scheduling	High — revenue model depends on this
Platform Admin	Monitors transactions, refunds, and disputes	Medium — financial audit and compliance

4.4 Risk Analysis

Risk	Likelihood	Impact	Mitigation
Payment gateway downtime	Medium	High	Support multiple gateways; graceful fallback UI
Double charge on retry	Medium	High	Idempotency keys on payment API; deduplication logic

Fraudulent transactions	Medium	High	Anomaly detection + OTP re-verification above threshold
Delayed vet payout	Medium	Medium	Automated payout schedule with manual override
Refund disputes	Medium	Medium	Clear refund policy in ToS; admin-managed dispute resolution
Sensitive payment data exposure	Low	High	PCI-DSS compliance; no raw card data stored

4.5 Detailed Flow

Step	Action	Metadata Used
1	Show Summary	AppointmentID, Amount
2	Select Method	PaymentMethod
3	Process Payment	TransactionID, Amount, PaymentMethod
4	Confirm Payment	PaymentStatus
5	Release Payout	VetID, TransactionID

5 Feature 5: Diagnosis & Prescription

5.1 Sub Features

- Consultation Conduct (In-Person / Zoom / Vet Visit)
- Diagnosis Entry
- Prescription & Dosage Entry
- Care Instructions Entry
- Follow-up Recommendation
- Diagnosis Submission to System
- Diagnosis Report Delivery to Farmer
- Disease Data Feed to Analytics Engine

5.2 Metadata

Consultation: AppointmentID, BookingType, ConsultationMode (In-Person / Zoom / VetVisit), ConsultationDate

Diagnosis: DiagnosisID, AppointmentID, VetID, FarmerID, AnimalType, DiseaseName, Severity

Prescription: PrescriptionID, DiagnosisID, Prescription, Dosage, DosageFrequency, DurationDays

Post-Consultation: CareInstructions, FollowupRequired, FollowupDate, ReportDeliveredAt

Analytics Feed: DiseaseID, DiseaseName, Area, Latitude, Longitude, ReportDate

5.3 Stakeholder Analysis

Stakeholder	Role / Interest	Impact Level
Veterinarian	Primary actor; conducts consultation, enters diagnosis, prescribes treatment and submits report	High — full clinical responsibility
Farmer / Pet Owner	Receives diagnosis report and prescription; follows care instructions and follow-up schedule	High — animal health outcome depends on this
VetCare Platform (Backend)	Stores DiagnosisID records; routes report to farmer; feeds disease data to analytics	High — central data persistence layer

Notification Engine	Delivers diagnosis report and follow-up reminders to the farmer's device	High — delivery failure means farmer misses treatment info
Disease Analytics / Heatmap Engine	Consumes submitted diagnosis data to update disease heatmap and risk scores	Medium — data dependency for Feature 6
Platform Admin	Audits diagnosis records for quality, compliance, and dispute resolution	Medium — medical liability and governance
Regulatory Authority	Ensures prescriptions comply with veterinary drug regulations and controlled substance laws	Medium — legal compliance
Pharmacy / Supplier (future)	May integrate to check drug availability or offer last-mile delivery of prescribed medicines	Low — potential future integration

5.4 Risk Analysis

Risk	Likelihood	Impact	Mitigation
Vet submits diagnosis without completing consultation	Medium	High	System enforces consultation status = "Completed" before diagnosis form unlocks
Incorrect dosage or drug name entry	Medium	High	Auto-suggest from formulary; mandatory confirmation step before save
Vet skips follow-up recommendation	High	Medium	FollowupRequired is a mandatory field; default FollowupDate pre-populated
Diagnosis report not delivered to farmer	Medium	High	Notification retry queue; in-app report accessible even if push fails

Diagnosis record tampering post-submission	Low	High	Immutable audit log for all DiagnosisID records; edits require admin approval
Disease data not fed to analytics engine	Low	Medium	Async event publish after DiagnosisID commit; dead-letter queue for failures
Prescription mis-used or shared outside platform	Medium	High	Prescriptions viewable only by linked FarmerID; watermarked PDF export
Regulatory non-compliance for controlled drugs	Low	High	Prescription drug list validated against approved veterinary formulary

5.5 Detailed Flow

Step	Action	Metadata Used	Actor
1	Open Appointment from Dashboard	AppointmentID, FarmerID	Veterinarian
2	Conduct Consultation (In-Person / Zoom / Vet Visit)	AppointmentID, BookingType, ConsultationMode	Vet + Farmer
3	Enter Diagnosis	DiseaseName, Severity, AnimalType	Veterinarian
4	Enter Prescription & Dosage	Prescription, Dosage, DosageFrequency, DurationDays	Veterinarian
5	Add Care Instructions	CareInstructions	Veterinarian
6	Set Follow-up Recommendation	FollowupRequired, FollowupDate	Veterinarian
7	Submit Diagnosis to System	DiagnosisID, AppointmentID, VetID	Veterinarian
8	System Stores Records	DiagnosisID, PrescriptionID, AppointmentID	Backend

9	Deliver Diagnosis Report to Farmer	FarmerID, Report-DeliveredAt, DiagnosisID	Notification Engine
10	Feed Disease Data to Analytics Engine	DiseaseID, DiseaseName, Area, Latitude, Longitude, ReportDate	Backend

6 Feature 6: Disease Heatmap

6.1 Sub Features

- Data Collection
- Risk Analysis
- Heatmap Generation

6.2 Metadata

DiseaseID, DiseaseName, AnimalType, Area, Latitude, Longitude, ReportDate, RiskScore

6.3 Stakeholder Analysis

Stakeholder	Role / Interest	Impact Level
Platform Admin	Manages heatmap data integrity and publication; triggers alerts	High — controls visibility of risk data
Farmer	Views disease risk in their area to make proactive decisions	High — animal safety depends on awareness
Veterinarian	Uses heatmap to anticipate local disease trends; adjusts readiness	Medium — informs clinical preparedness
Mapping / GIS Service	Provides underlying map tiles and GPS resolution	Medium — unavailability breaks map display

6.4 Risk Analysis

Risk	Likelihood	Impact	Mitigation
Insufficient data for accurate heatmap	High	High	Minimum report threshold before displaying risk zone
False alarm causing public panic	Medium	High	Admin approval gate before publishing high-risk zones
GPS coordinate errors	Medium	Medium	Area-level aggregation as fallback; outlier removal

6.5 Detailed Flow

Step	Action	Metadata Used
1	Collect Reports	DiseaseName, Area
2	Calculate Risk	RiskScore
3	Generate Heatmap	Latitude, Longitude, RiskScore
4	Show Map	Area, RiskScore

7 Feature 7: Farm Management

7.1 Sub Features

- Add Animal
- Edit Animal Status
- Livestock-wise Dashboard

7.2 Metadata

Animal: AnimalID, FarmerID, AnimalName, AnimalType, Breed, DateOfBirth, Age, HealthStatus, PregnancyStatus
Activity Record: ActivityID, Notes, RecordCreatedAt

7.3 Stakeholder Analysis

Stakeholder	Role / Interest	Impact Level
Farmer	Adds animals, manages animal profiles, and updates animal health status	High — directly depends on this feature for livestock management
Platform Admin	Monitors system activity and supports animal record management	Medium — ensures system reliability and data consistency

7.4 Risk Analysis

Risk	Likelihood	Impact	Mitigation
Incorrect animal information entry	Medium	Medium	Input validation and editable animal profile records
Incorrect animal status update	Medium	High	Editable audit logs for every status update
Animal data loss	Low	High	Regular database backup and record recovery mechanism

7.5 Detailed Flow

Step	Action	Metadata Used
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1	Add Animal	FarmerID, AnimalID, AnimalName, AnimalType, Breed, DateOfBirth, HealthStatus
2	Open Livestock Dashboard	FarmerID, AnimalID, AnimalName, AnimalType, HealthStatus
3	Select Animal Profile	AnimalID, AnimalName, Breed, Age, DateOfBirth
4	View Animal Description and Health Status	AnimalType, Breed, HealthStatus, PregnancyStatus, Notes
5	Edit Animal Status	AnimalID, HealthStatus, PregnancyStatus, Notes, RecordCreatedAt

8 Feature 8: Vaccination Management

8.1 Sub Features

- Calendar View for Upcoming Vaccines
- Vaccination Activity Recording
- Reminder and Due-Date Alert
- Vaccine Information View
- Vaccination History Tracking
- Vaccination Center Suggestion

8.2 Metadata

Vaccination: VaccinationID, AnimalID, FarmerID, VaccineName, DoseNumber, VaccinationDate, NextDueDate, VaccinationStatus, ReminderStatus

Vaccine Information: VaccineInfoID, DiseasePrevented, RecommendedAge, RecommendedInterval, SideEffects

Vaccination Center: CenterID, CenterName, CenterAddress, GPSLocation, Contact-Number

Activity Record: ActivityID, Notes, RecordCreatedAt

8.3 Stakeholder Analysis

Stakeholder	Role / Interest	Impact Level
Farmer	Checks vaccination schedules, receives reminders, and views vaccination history	High — directly depends on this feature for preventive livestock care
Vaccination Center	Provides vaccine availability, administers scheduled vaccines, and updates vaccination completion records	High — directly supports real-world vaccine delivery and service availability
Platform Admin	Maintains vaccine information, vaccination center data, and monitors system activity	Medium — ensures system reliability and data consistency
Notification Service	Sends vaccination reminders, due-date alerts, and overdue warnings	Medium — missed reminders reduce vaccination compliance
Location / Map Service	Suggests nearby vaccination centers based on farmer location	Medium — helps farmers access vaccination services easily

8.4 Risk Analysis

Risk	Likelihood	Impact	Mitigation
Missed vaccination schedule	High	High	Automated reminders through SMS, push notification, and dashboard alerts
Incorrect vaccination status update	Medium	High	Editable audit logs for every record update
Outdated vaccine information	Medium	Medium	Admin-maintained vaccine catalog with regular review
Poor vaccination center suggestion	Medium	Medium	Use GPS validation and allow manual filtering by area and vaccine availability
Reminder delivery failure	Medium	Medium	Use multiple channels such as SMS, push notification, and email fallback
Vaccination data loss	Low	High	Regular database backup and record recovery mechanism

8.5 Detailed Flow

Step	Action	Metadata Used
1	Create or View Vaccination Schedule	AnimalID, FarmerID, VaccineName, DoseNumber, RecommendedAge, NextDueDate
2	Show Upcoming Vaccines in Calendar	VaccinationID, VaccineName, NextDueDate, VaccinationStatus
3	Send Reminder Before Due Date	ReminderStatus, NextDueDate, FarmerID
4	Check Due Vaccine Information	VaccinationStatus, NextDueDate, VaccineName
5	View Vaccine Information	DiseasePrevented, RecommendedInterval, SideEffects

6	Suggest Nearby Vaccination Center	GPSLocation, CenterName, CenterAddress, ContactNumber
7	Record Vaccination Activity	VaccinationDate, CenterID, VaccineName, DoseNumber, Notes
8	Edit or Update Vaccination Status	VaccinationStatus, ActivityID, RecordCreatedAt
9	Store and Show Vaccination History	VaccinationID, VaccineName, VaccinationDate, NextDueDate, CenterID

9 Existing Solutions and Competitor Analysis

Several healthcare and veterinary service platforms currently exist; however, each provides only a subset of the features required for a complete livestock healthcare ecosystem.

- **Praava** (Human Healthcare): Provides appointment booking, diagnostic tests, health packages, video consultations, and promotional offers. However, it is designed for human healthcare and does not address livestock-specific requirements.
- **PranishabaVet**: Offers veterinary consultation services but lacks a polished user experience and advanced digital features.
- **Vetster**: Provides online veterinary consultations and service packages but lacks farm-centric monitoring and disease surveillance features.
- **Khamar Bondhu**: Includes disease detection support using a rule-based decision tree approach; however, its diagnostic capability is limited compared to modern AI-powered image analysis systems.

9.1 Our Innovation and Unique Selling Points

The proposed platform differentiates itself through the following innovations:

1. **Fully Functional Integrated Platform**

A complete ecosystem that combines appointment booking, veterinary consultation, disease diagnosis, farm monitoring, and health record management within a single platform.

2. **Affordable Service Model**

Designed to be cost-effective and accessible for small and medium-scale farmers.

3. **AI-Powered Disease Detection**

Utilizes deep learning models such as **MobileNet** and **EfficientNet** to identify livestock diseases from uploaded images, providing faster and more accurate preliminary diagnoses.

4. **Disease Heatmap and Outbreak Monitoring**

Introduces a geographical heatmap system to visualize disease occurrences and potential outbreak zones, enabling preventive action by farmers and veterinary authorities.

5. **Intelligent Chatbot Assistant (Proposed)**

An AI-driven chatbot capable of answering common farmer queries, providing basic guidance, and directing users to appropriate veterinary services.

9.2 Competitive Advantage

Compared to existing solutions, the proposed system combines veterinary consultation, AI-based disease detection, disease surveillance through heatmaps, and intelligent farmer assistance into a single affordable platform. This integrated approach aims to improve livestock healthcare accessibility while supporting early disease detection and outbreak prevention.